Annealed Multiplicative Closure at the Four–Möbius Gate: Prime–Sign Orthogonality, Product–Fiber Rigidity, and the Deterministic Walsh Corner

Liang Wang
School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

TPC–42 reduced the surviving triple–prime endpoint to a coherent orbit–slope energy containing one source and one target Möbius value per row. We deform all of those values by one common prime–sign field, rather than attaching independent row phases. For the zero alias, the source divisor and target affine value fuse into a squarefree integer label. The inherited scale separation $D = o(J)$ makes these labels injective across physical rows, and prime–sign orthogonality then makes the complete left–plus–right Hilbert energy equal its atomic diagonal in expectation, with no projective separation. For the full alias trace, the corresponding squarefree–kernel fibers have only $X^{o(1)}$ multiplicity; hence the annealed trace is still at the endpoint diagonal scale. Boolean hypercontractivity gives fixed high moments, and, for each fixed prescribed dyadic packet sequence, almost every global random multiplicative field closes every sufficiently large packet. At the distinguished all–minus prime corner the random model is exactly the Möbius function. We give an exact Walsh expansion there: every four–Möbius cross row collapses to one Möbius value of a squarefree symmetric–difference kernel. A sharp Walsh example shows that annealed diagonalization alone cannot control this specified corner. Thus the declared randomized multiplicative terminal model is closed, while the remaining deterministic problem is an explicit corner de–randomization theorem, not a claim of four–Möbius cancellation or a breach of sieve parity.

Keywords: Möbius correlation; random multiplicative function; Rademacher prime signs; product fiber; Walsh spectrum; Hilbert energy; triple–prime endpoint; parity boundary.

Convention. All complex inner products are linear in the first argument.

MSC 2020: Primary 11N35, 11K65; Secondary 42C10, 60E15, 11N36.

1 Introduction

The terminal TPC packet has reached a coefficient–specific arithmetic gate. TPC–41 closed the complete same–row alias sector and isolated a cross–row form whose literal summands contain

$$\mu(d_\alpha)\mu(d_\beta)\mu(m_\alpha j + h + kM)\mu(m_\beta j + h + k'M). \tag{1}$$

TPC–42 then proved that integers within one residue fiber are already orthogonal in the physical Hilbert coordinates. The remaining operation is coherent synthesis across distinct residue fibers, and projective mask separation reduces each layer to a weighted orbit–slope energy. Neither a scalar progression theorem nor a small localization boundary controls that synthesis [15, 16].

The present paper asks a narrower question before attempting the deterministic Möbius estimate:

> Does the endpoint fail for generic coefficient fields that preserve the same squarefree support and the same prime–level multiplicative coupling as the two physical Möbius occurrences?

The answer is no, in a strong exact sense. Choose one independent sign $\varepsilon_p \in \{-1, 1\}$ at every relevant prime and put

$$F_\varepsilon(n) = \mu^2(n) \prod_{p|n} \varepsilon_p. \tag{2}$$

We replace both $\mu(d_\alpha)$ and $\mu(m_\alpha j + h + kM)$ by this same function. This is not independent row noise. All occurrences sharing a prime share its sign, and the actual Möbius function is exactly the point $\varepsilon_p = -1$ for every prime.

1.1 Main results

The first result is a product–fiber theorem. For a nonzero atom, let

$$n_{\alpha,j,k} = \ell_\alpha d_\alpha j + h + kM, \qquad g_{\alpha,j,k} = (d_\alpha, n_{\alpha,j,k}), \tag{3}$$

and attach the squarefree Walsh label

$$s_{\alpha,j,k} = \frac{d_\alpha n_{\alpha,j,k}}{g_{\alpha,j,k}^2}. \tag{4}$$

At fixed (j, k, s), choosing $g \mid h + kM$ and $d \mid sg^2$ determines both $n = sg^2/d$ and

$$\ell = \frac{n - h - kM}{dj}. \tag{5}$$

Thus the fiber has divisor–type multiplicity $X^{o(1)}$. Walsh orthogonality, followed by Cauchy–Schwarz only inside these fibers, proves

$$\mathbb{E}_\varepsilon \sum_k \left(\|\mathscr{T}^L_{k,\varepsilon}\|^2 + \|\mathscr{T}^R_{k,\varepsilon}\|^2 \right) \ll X^{o(1)} \mathcal{D}_{\mathrm{tr}} \ll X^{o(1)} Q^2 J K_B. \tag{6}$$

This is the full physical trace endpoint in the annealed multiplicative model. It retains all deterministic masks and bypasses projective separation altogether.

The equality alias is exactly diagonal. There $(d_\alpha, n_{\alpha,j,0}) = 1$, so the label is the product

$$u_{\alpha,j} = d_\alpha(\ell_\alpha d_\alpha j + h). \tag{7}$$

If two rows share that product, then

$$j(\ell_\alpha d_\alpha^2 - \ell_\beta d_\beta^2) = h(d_\beta - d_\alpha). \tag{8}$$

The left side is either zero or has magnitude $\gg J$, whereas the right side is $O_h(D_0) = o(J)$. Hence the rows coincide. It follows that

$$\boxed{\mathbb{E}_\varepsilon \left(\|\mathscr{T}^L_{0,\varepsilon}\|^2 + \|\mathscr{T}^R_{0,\varepsilon}\|^2 \right) = \mathcal{D}_0.} \tag{9}$$

Since $K_B = X^{1/400+o(1)}$, all but a proportion $X^{-1/400+o(1)}$ of prime–sign environments meet the sufficient equality endpoint. Fixed Bonami–Beckner moments strengthen this to

$$\mathbb{P}_\varepsilon(Z_0 > K_B \mathcal{D}_0) \le X^{-q/400+o_q(1)} \tag{10}$$

for every fixed q. Coupling the signs across scales and applying Borel–Cantelli gives almost–sure eventual closure along any one fixed prescribed dyadic packet sequence.

The third result identifies exactly what annealing removes. For two zero–alias labels, put

$$\kappa_{\alpha,\beta,j} = \frac{u_{\alpha,j} u_{\beta,j}}{(u_{\alpha,j}, u_{\beta,j})^2}. \tag{11}$$

After grouping all deterministic masks and row representations into a coefficient $C(\kappa)$, the physical energy is

$$\boxed{\|\mathscr{T}_0^L\|^2 + \|\mathscr{T}_0^R\|^2 = \mathcal{D}_0 + \sum_{\kappa>1} \mu(\kappa) C(\kappa).} \tag{12}$$

This uses the literal identity

$$\mu(d_\alpha)\mu(n_\alpha)\mu(d_\beta)\mu(n_\beta) = \mu(\kappa_{\alpha,\beta,j}). \tag{13}$$

Thus the four signs have been compressed to one Möbius value on a squarefree symmetric–difference spectrum. Randomizing only primes up to z deletes exactly those terms whose kernel has a prime factor at most z, leaving an exact rough–kernel filtration.

Finally, the annealed–to–corner distinction is sharp. On the divisor Walsh system of a squarefree product with R divisors, one may have mean energy R and all–minus energy R^2, with the polynomial vanishing at every other corner. Hence a small exceptional set can be concentrated entirely at the physical phase. No measure selection can replace a deterministic theorem for (12).

1.2 What has and has not moved

The results close the generic multiplicative geometry, not the physical Möbius point. In particular, they prove that the following features do not by themselves force the J^2 coherent loss:

(i) squarefree support;

(ii) one globally coupled multiplicative sign field at source and target;

(iii) the actual row masks and opposite–row Hilbert coordinates;

(iv) the complete alias trace; and

(v) the endpoint aspect ratios of the triple–prime packet.

What remains is a quenched theorem excluding the all–minus prime point from a small, potentially structured exceptional set. We do not call (6) a Möbius correlation estimate: for generic prime signs, $F_\varepsilon * \mathbf{1} = \delta_1$ is false, and the actual point cannot be selected probabilistically.

1.3 Organization

Section 2 records the exact inherited terminal object. Section 3 defines the consistent multiplicative deformation. Section 4 proves the divisor fiber bound, and section 5 applies it to the full trace. Section 6 proves singleton fibers at equality. Section 7 gives hypercontractive and almost–sure extensions. Section 8 derives the exact physical kernel spectrum, while section 9 proves its sharp selection obstruction. The final sections compare adjacent literature, state the next deterministic gate, and delimit the exact certificate.

2 The inherited physical interface

We use only the terminal statements proved in TPC–18, TPC–36, TPC–41, and TPC–42 [12, 13, 15, 16]. Fix $h \in \mathbb{Z} \setminus \{0\}$. A source row is

$$\alpha = (\ell_\alpha, d_\alpha), \qquad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \qquad \ell_\alpha \asymp L_0, \qquad d_\alpha \asymp D_0, \tag{14}$$

where ℓ_α is prime, d_α is squarefree, $(d_\alpha, h) = 1$, and $L_0 D_0 = Q$. Source separation makes m_α determine α. The orbit variable belongs to one of the inherited dyadic blocks

$$a_J J \le j \le b_J J, \qquad QJ \asymp X, \tag{15}$$

for fixed positive $a_J < b_J$. At the endpoint,

$$Q = X^{267/400+o(1)}, \qquad J = X^{133/400+o(1)}, \tag{16}$$

$$D_0 = X^{10049/52500+o(1)} = o(J). \tag{17}$$

In particular, for all sufficiently large X,

$$\min \mathcal{J} > |h| \max_\alpha d_\alpha. \tag{18}$$

Let M be the inherited triple–prime modulus and let $\mathcal{I}_B = \{-L_B, \dots, L_B\}$, where

$$K_B := |\mathcal{I}_B| = X^{1/400+o(1)}, \qquad K_B = X^{o(1)} \frac{Q}{J^2}. \tag{19}$$

For $k \in \mathcal{I}_B$, put

$$c_k = h + kM, \qquad n_{\alpha,j,k} = m_\alpha j + c_k. \tag{20}$$

Because h is fixed and nonzero while $M > |h|$ for all sufficiently large X, every c_k occurring here is nonzero: this is immediate for $k = 0$, and $|kM| > |h|$ for $k \neq 0$. All terminal coefficients are interpreted as zero outside the declared positive support. Uniformly on that support,

$$|c_k| + d_\alpha n_{\alpha,j,k} \le X^{C_0} \tag{21}$$

for one fixed C_0.

The source coefficient factors as

$$\gamma_\alpha^{(i)} = \mu(d_\alpha)\gamma_\alpha^{\circ,(i)}, \qquad i = 1, 2, \tag{22}$$

where $\gamma^{\circ,(i)}$ contains the source logarithm, smooth weight, and fixed residue data, but no Möbius sign. After also stripping the target Möbius sign, define deterministic coefficients

$$a^L_{\alpha,\gamma,j,k} := -\gamma_\alpha^{\circ,(1)} \mathfrak{A}^{\mathrm{act}}_{\alpha,\gamma}(j) \log(n_{\alpha,j,k}) \Phi_0(n_{\alpha,j,k}), \tag{23}$$

again set to zero outside the terminal support. Then the physical left column is exactly

$$[\mathscr{T}_k^L]_{\gamma,j} = \sum_\alpha a^L_{\alpha,\gamma,j,k} \mu(d_\alpha) \mu(n_{\alpha,j,k}). \tag{24}$$

The right column has coefficients $a^R_{\alpha,\gamma,j,k}$ obtained by exchanging the two row systems. Conditional on the inherited row geometry and terminal support, no additional structural assumption is made on the deterministic complex values $a^{L/R}$; in particular, they are independent of the prime–sign environment.

Define the atomic diagonals

$$\mathcal{D}_{k,L} := \sum_{\gamma,j,\alpha} |a^L_{\alpha,\gamma,j,k}|^2 \mu^2(n_{\alpha,j,k}), \tag{25}$$

$$\mathcal{D}_0 := \mathcal{D}_{0,L} + \mathcal{D}_{0,R}, \qquad \mathcal{D}_{\mathrm{tr}} := \sum_{k\in\mathcal{I}_B} (\mathcal{D}_{k,L} + \mathcal{D}_{k,R}). \tag{26}$$

TPC–41 and TPC–42 give

$$\mathcal{D}_0 \ll X^{o(1)} Q^2 J, \tag{27}$$

$$\mathcal{D}_{\mathrm{tr}} \ll X^{o(1)} Q^2 J K_B. \tag{28}$$

The physical equality target is

$$\|\mathscr{T}_0^L\|^2 + \|\mathscr{T}_0^R\|^2 \ll X^{o(1)} K_B Q^2 J, \tag{29}$$

while the stronger trace target is

$$\sum_{k\in\mathcal{I}_B} \left(\|\mathscr{T}_k^L\|^2 + \|\mathscr{T}_k^R\|^2\right) \ll X^{o(1)} Q^2 J K_B. \tag{30}$$

These two estimates are not assumed.

3 One multiplicative prime–sign field

Let $(\varepsilon_p)_{p\in\mathcal{P}}$ be independent signs, uniformly distributed on $\{-1,1\}$, for every prime occurring in the finite packet. Extend the prime signs to the completely multiplicative phase $\varepsilon(n) = \prod_p \varepsilon_p^{v_p(n)}$, and define the squarefree–supported random multiplicative function

$$F_\varepsilon(n) := \mu^2(n) \prod_{p|n} \varepsilon_p. \tag{31}$$

Thus F_ε is multiplicative on coprime inputs and has the same zero set and absolute values as μ. It is not called completely multiplicative because its value is zero on nonsquarefree integers. At the distinguished prime point

$$\varepsilon_p = -1 \quad (p \in \mathcal{P}), \tag{32}$$

one has

$$F_\varepsilon(n) = \mu(n) \tag{33}$$

for every relevant integer.

The deformation is applied consistently to the two Möbius occurrences in each physical row. Define

$$[\mathscr{T}^L_{k,\varepsilon}]_{\gamma,j} := \sum_\alpha a^L_{\alpha,\gamma,j,k} F_\varepsilon(d_\alpha) F_\varepsilon(n_{\alpha,j,k}), \tag{34}$$

and define $\mathscr{T}^R_{k,\varepsilon}$ symmetrically. In particular,

$$\mathscr{T}^{L/R}_{k,-\mathbf{1}} = \mathscr{T}^{L/R}_k. \tag{35}$$

This is not an independent row randomization: the same prime sign appears at every source and target integer divisible by that prime.

For a squarefree positive integer s, write

$$\chi_s(\varepsilon) := \prod_{p|s} \varepsilon_p, \qquad \chi_1 = 1. \tag{36}$$

These are the Walsh characters of the finite prime cube and satisfy

$$\mathbb{E}_\varepsilon \chi_s(\varepsilon)\chi_t(\varepsilon) = \mathbf{1}_{s=t}. \tag{37}$$

If both d and n are squarefree, put

$$g = (d,n), \qquad s(d,n) := \frac{dn}{g^2}. \tag{38}$$

Then $s(d,n)$ is squarefree and

$$F_\varepsilon(d)F_\varepsilon(n) = \chi_{s(d,n)}(\varepsilon). \tag{39}$$

If n is not squarefree, both sides of every physical atom in which it occurs are interpreted as zero. Therefore no conditioning on squarefreeness is hidden in the model.

Remark 3.1 (Relation to external row phases). Independent signs attached directly to row labels would diagonalize any finite row family and would be a generic phase–frame statement of the kind already delimited in TPC–40. In (34), the signs live at primes, respect multiplication, occur simultaneously at the source and target, and have the actual Möbius function as the point (32). The product–fiber geometry needed to control their collisions is the arithmetic content of the next section.

4 Squarefree–kernel fibers for all aliases

Fix (j,k) and abbreviate $c = c_k$. For a row whose random atom is nonzero, both d_α and $n_{\alpha,j,k} = \ell_\alpha d_\alpha j + c$ are squarefree. Define

$$g_{\alpha,j,k} := (d_\alpha, n_{\alpha,j,k}), \qquad s_{\alpha,j,k} := \frac{d_\alpha n_{\alpha,j,k}}{g_{\alpha,j,k}^2}. \tag{40}$$

Since $d_\alpha \mid \ell_\alpha d_\alpha j$,

$$g_{\alpha,j,k} = (d_\alpha, c_k). \tag{41}$$

Lemma 4.1 (Divisor control of one kernel fiber)**.** *For fixed (j,k,s), the number $r_{j,k}(s)$ of physical rows satisfying $s_{\alpha,j,k} = s$ obeys*

$$r_{j,k}(s) \le \sum_{g \mid |c_k|} \tau(sg^2). \tag{42}$$

Consequently there is a constant C_1, depending only on the fixed support exponents, such that

$$\rho_X := \max_{j,k,s} r_{j,k}(s) \le \exp\left(C_1 \frac{\log X}{\log\log X}\right) = X^{o(1)}. \tag{43}$$

Proof. If a row belongs to the s-fiber and $g = g_{\alpha,j,k}$, then

$$g \mid c_k, \qquad d_\alpha n_{\alpha,j,k} = sg^2. \tag{44}$$

Fix such a divisor g and put $R = sg^2$. A choice $d_\alpha \mid R$ determines

$$n_{\alpha,j,k} = R/d_\alpha, \qquad \ell_\alpha = \frac{R/d_\alpha - c_k}{d_\alpha j}. \tag{45}$$

There is at most one row for that divisor; integrality, primality, dyadic support, and the physical masks can only delete choices. Hence there are at most $\tau(sg^2)$ rows for fixed g, proving (42).

By (21), every integer in the divisor functions is at most a fixed power of X. The standard uniform divisor bound gives [7, Chapter I]

$$\max_{n \le X^{O(1)}} \tau(n) \le \exp\left(O\left(\frac{\log X}{\log\log X}\right)\right). \tag{46}$$

The same estimate applies to the number of divisors of $|c_k|$. Absorbing the two constants proves (43). □

Remark 4.2 (Why the alias case is not injective). For $k \neq 0$, neither $(d_\alpha, c_k) = 1$ nor singleton kernel fibers are assumed. The proof retains the common prime factors through g^2 and pays their exact divisor multiplicity. The sharper singleton statement at $k = 0$ uses the fixed intercept h and the scale separation $D_0 = o(J)$; see section 6.

5 Annealed closure of the full physical trace

For fixed (γ, j, k), group the left coefficients by their squarefree–kernel label:

$$A^L_{\gamma,j,k}(s) := \sum_{\alpha : s_{\alpha,j,k} = s} a^L_{\alpha,\gamma,j,k} \mu^2(n_{\alpha,j,k}). \tag{47}$$

Equation (39) gives the exact Walsh polynomial

$$[\mathscr{T}^L_{k,\varepsilon}]_{\gamma,j} = \sum_s A^L_{\gamma,j,k}(s) \chi_s(\varepsilon). \tag{48}$$

Theorem 5.1 (Full trace at the annealed diagonal scale)**.** *For the complete physical alias family,*

$$\begin{aligned} \mathbb{E}_\varepsilon \sum_{k \in \mathcal{I}_B} \big(\|\mathscr{T}^L_{k,\varepsilon}\|^2 + \|\mathscr{T}^R_{k,\varepsilon}\|^2 \big) \\ \leq \rho_X \mathcal{D}_{\mathrm{tr}} \ll X^{o(1)} Q^2 J K_B. \end{aligned} \tag{49}$$

The estimate retains the actual masks, orbit coordinates, source weights, terminal cutoffs, and all aliases. It does not use the projective layer decomposition.

Proof. Walsh orthogonality gives

$$\mathbb{E}_\varepsilon |[\mathscr{T}^L_{k,\varepsilon}]_{\gamma,j}|^2 = \sum_s |A^L_{\gamma,j,k}(s)|^2. \tag{50}$$

For one fiber, Cauchy–Schwarz and lemma 4.1 give

$$\begin{aligned} |A^L_{\gamma,j,k}(s)|^2 &\leq r_{j,k}(s) \sum_{\alpha : s_{\alpha,j,k} = s} |a^L_{\alpha,\gamma,j,k}|^2 \mu^2(n_{\alpha,j,k}) \\ &\leq \rho_X \sum_{\alpha : s_{\alpha,j,k} = s} |a^L_{\alpha,\gamma,j,k}|^2 \mu^2(n_{\alpha,j,k}). \end{aligned} \tag{51}$$

Sum over s, γ, j, k to obtain the left contribution $\rho_X \sum_k \mathcal{D}_{k,L}$. The same argument gives the right contribution. Now use (43) and the inherited diagonal bound (28). □

Within the annealed multiplicative model, the theorem removes the full factor

$$\frac{Q}{K_B} = X^{o(1)} J^2 \tag{52}$$

that separates the deterministic overlap estimate from the sufficient equality endpoint in TPC–42. The statement is annealed: it concerns a single globally multiplicative prime–sign field averaged over its prime coordinates, not the distinguished all–minus point.

The same geometry acts directly on the minimal folded bank. Let $\mathcal{K}_0 = \{0\}$, and for $1 \leq a \leq L_B$ let $\mathcal{K}_a = \{a, a - (L_B + 1)\}$. Define

$$[\mathscr{U}^L_{a,\varepsilon}]_{\gamma,j} := \sum_\alpha \sum_{k \in \mathcal{K}_a} a^L_{\alpha,\gamma,j,k} F_\varepsilon(d_\alpha) F_\varepsilon(n_{\alpha,j,k}), \tag{53}$$

and define the right side symmetrically.

Corollary 5.2 (Annealed closure of the actual minimal bank)**.** *One has*

$$\mathbb{E}_\varepsilon \sum_{a=0}^{L_B} \big(\|\mathscr{U}^L_{a,\varepsilon}\|^2 + \|\mathscr{U}^R_{a,\varepsilon}\|^2\big) \le 2\rho_X \mathcal{D}_{\mathrm{tr}} \ll X^{o(1)} Q^2 J K_B. \tag{54}$$

Proof. At fixed (γ, j, a), expand the inner sum into atoms (α, k) with $k \in \mathcal{K}_a$. For each Walsh label there are at most two alias choices, and for each alias there are at most ρ_X rows. Walsh orthogonality and fiberwise Cauchy–Schwarz therefore cost at most $2\rho_X$. Summing the raw atomic coefficients over the disjoint buckets gives exactly $\mathcal{D}_{\mathrm{tr}}$. $\square$

Corollary 5.3 (One–scale typical trace closure)**.** *Let $L_X \to \infty$ be any function. Then*

$$\mathbb{P}_\varepsilon\left(\sum_k \big(\|\mathscr{T}^L_{k,\varepsilon}\|^2 + \|\mathscr{T}^R_{k,\varepsilon}\|^2\big) > L_X \rho_X \mathcal{D}_{\mathrm{tr}}\right) \le L_X^{-1}. \tag{55}$$

In particular, choosing any slowly growing $L_X \to \infty$ with $L_X = X^{o(1)}$ yields the trace endpoint for all but an $o(1)$ proportion of prime–sign environments.

Proof. If $\mathcal{D}_{\mathrm{tr}} = 0$, the annealed bound forces the trace energy to vanish almost surely. Otherwise apply Markov's inequality to that nonnegative energy and use theorem 5.1. $\square$

Remark 5.4 (A finite exact ensemble). Only primes dividing the finitely many source and terminal integers enter the packet. Thus every expectation above is a normalized sum over a finite cube. No limiting probability space or independence between different values of $F_\varepsilon(n)$ is being assumed.

6 Exact fused–label rigidity at the equality alias

At $k = 0$, the intercept is the fixed integer h. Hence

$$(d_\alpha, n_{\alpha,j,0}) = (d_\alpha, h) = 1, \tag{56}$$

and every nonzero random atom is indexed by the squarefree product

$$u_{\alpha,j} := d_\alpha n_{\alpha,j,0} = d_\alpha(\ell_\alpha d_\alpha j + h). \tag{57}$$

Lemma 6.1 (Fused–label injectivity)**.** *For all sufficiently large X, fixed $j \in \mathcal{J}$, and two physical rows α, β,*

$$u_{\alpha,j} = u_{\beta,j} \quad\Longrightarrow\quad \alpha = \beta. \tag{58}$$

Proof. Equality of the two labels gives

$$j(\ell_\alpha d_\alpha^2 - \ell_\beta d_\beta^2) = h(d_\beta - d_\alpha). \tag{59}$$

If the integer in parentheses on the left is nonzero, the left side has absolute value at least j, whereas

$$|h(d_\beta - d_\alpha)| \le |h| \max_\rho d_\rho < j \tag{60}$$

by (18). This is impossible. Thus the parenthesized integer vanishes, and (59), together with $h \neq 0$, forces $d_\alpha = d_\beta$. It then forces $\ell_\alpha = \ell_\beta$, so the rows coincide. $\square$

The use of $j \asymp J$ is essential. Merely knowing $|\mathcal{J}| = O(J)$ would not prove the strict inequality in (60).

Theorem 6.2 (Exact equality–column diagonalization)**.** *Put*

$$Z_0(\varepsilon) := \|\mathscr{T}^L_{0,\varepsilon}\|^2 + \|\mathscr{T}^R_{0,\varepsilon}\|^2. \tag{61}$$

Then

$$\boxed{\mathbb{E}_\varepsilon Z_0(\varepsilon) = \mathcal{D}_0.} \tag{62}$$

Consequently, for every $\lambda > 0$,

$$\mathbb{P}_\varepsilon\big(Z_0(\varepsilon) > \lambda \mathcal{D}_0\big) \le \lambda^{-1}. \tag{63}$$

At the inherited endpoint,

$$\mathbb{P}_\varepsilon\big(Z_0(\varepsilon) > K_B \mathcal{D}_0\big) \le K_B^{-1} = X^{-1/400+o(1)}. \tag{64}$$

Thus all but at most a proportion $X^{-1/400+o(1)}$ of the multiplicative prime–sign environments satisfy the random–model equality benchmark at the same numerical scale as (29). This statement does not select the physical all–minus environment.

Proof. For fixed (γ, j), the nonzero row atoms have distinct Walsh labels $u_{\alpha,j}$ by lemma 6.1. Equation (37) therefore deletes every unequal row pair and leaves

$$\mathbb{E}_\varepsilon |[\mathscr{T}^L_{0,\varepsilon}]_{\gamma,j}|^2 = \sum_\alpha |a^L_{\alpha,\gamma,j,0}|^2 \mu^2(n_{\alpha,j,0}). \tag{65}$$

Sum over the left coordinates and repeat on the right to obtain (62). The probability statements are immediate if $\mathcal{D}_0 = 0$, since then the nonnegative variable Z_0 vanishes almost surely. Otherwise they are Markov's inequality, followed by (19) and the inherited diagonal bound. □

7 Hypercontractive moments and almost–sure dyadic closure

The first moment is enough for the endpoint statements above. Fixed higher moments give stronger tail bounds for a random prime–sign environment, while still leaving open exceptional behavior at the specified all–minus corner. We use the standard Bonami–Beckner inequality on the discrete cube [2, 3]; the weighted random multiplicative form needed below is also recorded explicitly by Harper [4, Probability Result 2.3].

Lemma 7.1 (One coordinate)**.** *Let*

$$S(\varepsilon) = \sum_s A_s \chi_s(\varepsilon) \tag{66}$$

be any coordinate polynomial produced by the physical packet. For each fixed integer $q \ge 1$,

$$\|S\|^2_{L^{2q}} \le \sum_s (2q-1)^{\omega(s)} |A_s|^2 \le X^{o_q(1)} \sum_s |A_s|^2. \tag{67}$$

Proof. The first inequality is Bonami–Beckner hypercontractivity, decomposed by Walsh degree. Every label in the packet is at most $X^{O(1)}$, so the standard maximal-order bound gives

$$\max_s \omega(s) \ll \frac{\log X}{\log\log X}. \tag{68}$$

For fixed q, the factor $(2q-1)^{\omega(s)}$ is therefore $X^{o_q(1)}$. □

Theorem 7.2 (Fixed moments of the physical energies)**.** *For every fixed integer $q \geq 1$,*

$$\mathbb{E}_\varepsilon Z_0(\varepsilon)^q \leq X^{o_q(1)} \mathcal{D}_0^q, \tag{69}$$

$$\mathbb{E}_\varepsilon Z_{\mathrm{tr}}(\varepsilon)^q \leq X^{o_q(1)} \mathcal{D}_{\mathrm{tr}}^q, \tag{70}$$

where

$$Z_{\mathrm{tr}}(\varepsilon) := \sum_k \big(\| \mathscr{T}_{k,\varepsilon}^L \|^2 + \| \mathscr{T}_{k,\varepsilon}^R \|^2 \big). \tag{71}$$

In particular,

$$\mathbb{P}_\varepsilon \big(Z_0(\varepsilon) > K_B \mathcal{D}_0 \big) \leq X^{-q/400 + o_q(1)}. \tag{72}$$

Proof. For each output coordinate i, write $S_i = \sum_s A_{i,s} \chi_s$ and let D_i be its atomic diagonal. The fiber Cauchy estimate (51) gives

$$\sum_s |A_{i,s}|^2 \leq \rho_X D_i. \tag{73}$$

For $k = 0$, one may replace ρ_X by 1. By lemma 7.1,

$$\big(\mathbb{E} |S_i|^{2q} \big)^{1/q} = \| S_i \|_{2q}^2 \leq X^{o_q(1)} D_i. \tag{74}$$

Minkowski's inequality in L^q now gives

$$\big(\mathbb{E} Z^q \big)^{1/q} = \left\| \sum_i |S_i|^2 \right\|_{L^q} \leq \sum_i \| |S_i|^2 \|_{L^q} \leq X^{o_q(1)} \sum_i D_i. \tag{75}$$

This proves both moment estimates. Apply Markov with threshold $K_B^q \mathcal{D}_0^q$ when $\mathcal{D}_0 > 0$; if $\mathcal{D}_0 = 0$, the first moment already gives $Z_0 = 0$ almost surely. Finally use $K_B = X^{1/400 + o(1)}$ to obtain (72). □

The field can be coupled across scales by choosing one sign for every prime once and for all.

Corollary 7.3 (Almost–sure closure on prescribed dyadic packets)**.** *Let $X_r = 2^r$, and for each r fix one deterministic inherited TPC packet. For almost every global prime–sign field, all sufficiently large r satisfy*

$$Z_0(X_r, \varepsilon) \leq K_B(X_r) \mathcal{D}_0(X_r), \tag{76}$$

$$Z_{\mathrm{tr}}(X_r, \varepsilon) \leq X_r^{o(1)} \mathcal{D}_{\mathrm{tr}}(X_r). \tag{77}$$

Consequently both random–model endpoint estimates hold eventually on that dyadic packet sequence.

Proof. The failure probabilities in (64) are summable at $X_r = 2^r$. For the trace, use (43) and set

$$L_X = \exp\!\left(\frac{\log X}{\sqrt{\log \log X}} \right) = X^{o(1)}. \tag{78}$$

For sufficiently large X, Markov and theorem 5.1 give a failure probability at most

$$\mathbb{P}_\varepsilon \big(Z_{\mathrm{tr}} > L_X \mathcal{D}_{\mathrm{tr}} \big) \leq \frac{\rho_X}{L_X} \leq \exp\!\left(-c \frac{\log X}{\sqrt{\log \log X}} \right), \tag{79}$$

which is summable on the dyadic sequence. The first Borel–Cantelli lemma proves both assertions; no independence between scales is needed. □

The packet sequence in corollary 7.3 is prescribed. Uniformity over an additional family of packets requires an explicit entropy or cardinality ledger and is not asserted.

8 The deterministic Möbius corner as one Walsh spectrum

The equality alias admits a sharper deterministic description than the annealed theorem. For nonzero atoms, let $u_{\alpha,j}$ be the squarefree fused label in (57). For two rows at the same orbit time, define their squarefree symmetric–difference kernel

$$\kappa_{\alpha,\beta,j} := \frac{u_{\alpha,j} u_{\beta,j}}{(u_{\alpha,j}, u_{\beta,j})^2}. \tag{80}$$

It is squarefree and

$$\chi_{u_{\alpha,j}}(\varepsilon)\chi_{u_{\beta,j}}(\varepsilon) = \chi_{\kappa_{\alpha,\beta,j}}(\varepsilon). \tag{81}$$

For $\kappa > 1$, let $C(\kappa)$ be the sum of all ordered unequal row coefficients, on both physical sides, whose pair kernel is κ:

$$\begin{aligned} C(\kappa) := \sum_{\gamma,j} \sum_{\substack{\alpha \neq \beta \\ \kappa_{\alpha,\beta,j} = \kappa}} & a^L_{\alpha,\gamma,j,0} \overline{a^L_{\beta,\gamma,j,0}} \mu^2(n_{\alpha,j,0}) \mu^2(n_{\beta,j,0}) \\ & + \text{the symmetric right contribution.} \end{aligned} \tag{82}$$

Pairing (α, β) with (β, α) shows that $C(\kappa) \in \mathbb{R}$.

Theorem 8.1 (Exact four–to–one Möbius compression)**.** *The complete equality energy has the Walsh expansion*

$$\boxed{Z_0(\varepsilon) = \mathcal{D}_0 + \sum_{\kappa > 1} C(\kappa) \chi_\kappa(\varepsilon).} \tag{83}$$

At the physical all–minus point,

$$\boxed{\|\mathscr{T}_0^L\|^2 + \|\mathscr{T}_0^R\|^2 = \mathcal{D}_0 + \sum_{\kappa > 1} \mu(\kappa) C(\kappa).} \tag{84}$$

Moreover,

$$\mathbb{E}_\varepsilon |Z_0(\varepsilon) - \mathcal{D}_0|^2 = \sum_{\kappa > 1} |C(\kappa)|^2. \tag{85}$$

Proof. Expand the two squared Hilbert norms defining Z_0. By (81), every ordered row pair contributes the Walsh character indexed by its symmetric–difference kernel. The constant character occurs exactly when $u_{\alpha,j} = u_{\beta,j}$, which by lemma 6.1 means $\alpha = \beta$. Its coefficient is therefore $\mathcal{D}_0$, proving (83).

At $\varepsilon_p = -1$,

$$\chi_\kappa(-\mathbf{1}) = (-1)^{\omega(\kappa)} = \mu(\kappa). \tag{86}$$

For a contributing pair, this also gives the literal identity

$$\begin{aligned} &\mu(d_\alpha)\mu(n_{\alpha,j,0})\mu(d_\beta)\mu(n_{\beta,j,0}) \\ &\qquad = \mu(u_{\alpha,j})\mu(u_{\beta,j}) = \mu(\kappa_{\alpha,\beta,j}). \end{aligned} \tag{87}$$

Thus (84) is precisely the physical four–Möbius expansion, not a surrogate. Finally, Walsh orthogonality and the absence of a nonzero constant term give (85). □

The theorem replaces a row–pair family of four Möbius signs by one signed spectral sum over squarefree kernels. The coefficient $C(\kappa)$ still contains all masks and all representations of the same kernel; no cancellation estimate for that sum is asserted.

There is an exact small–prime filtration. Fix $z \geq 2$, average the signs ε_p for $p \leq z$, and set $\varepsilon_p = -1$ for $p > z$. Let $P^-(n)$ denote the least prime factor of $n > 1$.

Corollary 8.2 (Rough–kernel filtration)**.** *One has*

$$\boxed{\mathbb{E}_{\varepsilon_p:\,p\le z} Z_0(\varepsilon_{\le z}, -\mathbf{1}_{>z}) = \mathcal{D}_0 + \sum_{\substack{\kappa>1\\ P^-(\kappa)>z}} \mu(\kappa)C(\kappa).} \tag{88}$$

Proof. The average of χ_κ is zero if any prime $p \le z$ divides κ. Otherwise every prime in κ keeps the value -1, and the character equals $\mu(\kappa)$. Insert this into (83). □

Before any relevant prime is averaged one has the physical corner; after every relevant prime has been averaged, only $\mathcal{D}_0$ remains. A future de–randomization may therefore attack the successive prime increments or the final rough spectrum. The identity itself does not bound either.

9 Sharp annealed–to–corner boundaries

The exceptional set in an annealed theorem can contain the specified all–minus point. This is not a technical possibility: Walsh systems admit exact extremizers.

Theorem 9.1 (A one–corner extremizer)**.** *Let $P = p_1 \cdots p_s$ be squarefree and let $\mathcal{U} = \{u : u \mid P\}$, so $R := |\mathcal{U}| = 2^s$. Define*

$$Y(\varepsilon) := \sum_{u\mid P} \mu(u)\chi_u(\varepsilon). \tag{89}$$

Then the labels are distinct and orthogonal,

$$Y(\varepsilon) = \prod_{p\mid P}(1-\varepsilon_p), \tag{90}$$

$$\mathbb{E}|Y|^2 = R, \tag{91}$$

$$|Y(-\mathbf{1})|^2 = R^2, \tag{92}$$

and Y vanishes at every other point of the s-dimensional sign cube. Thus the corner–to–annealed energy ratio is exactly R, and Markov's inequality is sharp on this multiplicative Walsh class.

Proof. Expanding $\prod_{p\mid P}(1-\varepsilon_p)$ gives (89). The orthogonality of its R Walsh characters gives $\mathbb{E}|Y|^2 = R$. At the all–minus point every factor is two, so $Y(-\mathbf{1}) = 2^s = R$. At any other point one factor vanishes. □

The same ratio is obtained from any R pairwise distinct squarefree labels of the same Möbius parity, with equal unit coefficients in one coherent output coordinate. In particular, the fused–label injectivity of lemma 6.1 does not by itself control the physical corner; it is exactly the property that makes the annealed constant term diagonal.

Remark 9.2 (No universal row theorem)*.* The physical orbit matrix has only $O(J)$ output times and polynomially many source rows. A Bessel inequality required for every row coefficient direction must therefore pay the usual rank–trace lower bound. This is the dimension obstruction already isolated in TPC–40, and is why TPC–42 restricted its proposed orbit–slope theorem to the actual projective directions. The present random theorem goes further in a different direction: it retains the complete physical coefficient field and proves that almost every consistent multiplicative prime environment avoids the coherent mode.

Remark 9.3 (What the model does not preserve)*.* For generic ε, one has $F_\varepsilon * \mathbf{1} \ne \delta_1$; only the distinguished all–minus point recovers $F_{-\mathbf{1}} = \mu$ and hence the Möbius inversion identity $\mu * \mathbf{1} = \delta_1$. Therefore theorem 5.1 is a theorem for the randomized terminal interface, not a randomized proof about primes. It preserves the exact squarefree support, multiplicative coupling, masks, aliases, and Hilbert geometry that occur at the final gate; it deliberately does not preserve every special algebraic identity of the distinguished Möbius point.

10 Literature comparison and method boundary

Random multiplicative functions have a substantial literature. In the standard Rademacher model the prime values are independent signs and the function is supported on squarefree integers; in the Steinhaus model the prime values are independent Haar phases. Their partial sums and moments exhibit behavior very different from sums of independent integer–indexed variables; see, for example, Harper [4, 5], Heap and Lindqvist [6]. The present theorem is not a new general moment theorem for those models. Its new input is the TPC–specific product–fiber calculation in lemma 4.1, which permits one globally coupled prime field to pass through the physical source and target simultaneously. For integral q, Probability Result 2.3 of Harper [4] gives the exact weighted inequality

$$\mathbb{E}\left|\sum_{n\le N} a_n F_\varepsilon(n)\right|^{2q} \le \left(\sum_{n\le N} |a_n|^2 d_{2q-1}(n)\right)^q . \tag{93}$$

Here d_r denotes the r-fold divisor function. On a squarefree kernel s, $d_{2q-1}(s) = (2q-1)^{\omega(s)}$, so this is exactly the coordinate estimate used in lemma 7.1. Its underlying cube inequality goes back to Beckner [2], Bonami [3]. The subsequent Hilbert–energy estimate is packet specific and follows by fiberwise Cauchy–Schwarz and Minkowski; no asymptotic theorem for unweighted random partial sums is imported.

10.1 Why this is not another external phase bank

TPC–39 used additive alias phases to recover the equality bucket, and TPC–40 proved a sharp recovery–decorrelation uncertainty principle for external row and alias frames [14]. An independent sign on each row, or a complete Dirichlet–character average separating distinct row residues, would give an immediate diagonal second moment. Such a calculation is generic linear algebra and cannot recover the prescribed untwisted column at low amplification.

Here the deformation instead replaces the two actual Möbius occurrences by the same squarefree–supported multiplicative field. Prime factors shared by different source and target integers therefore create genuine dependencies. The divisor fibers in (42), and their exact collapse at $k = 0$, are what make the annealed theorem work. No recovery map is applied, and $F_{-\mathbf{1}} = \mu$ is one member of the coefficient family. This places the result on the coefficient–specific side of the boundary left open by TPC–40.

10.2 Adjacent deterministic multiplicative correlations

There are powerful deterministic results for averaged or fixed–complexity systems of multiplicative functions. The theorem of Matomäki et al. [8] averages fixed–order correlations over their additive shifts, including the two–point case. Tao proved the logarithmically averaged two–point Chowla and Elliott statements for fixed affine forms [11]. Higher uniformity in short intervals on average and applications to averaged polynomial patterns were established by Matomäki et al. [9]. More recently, Matthiesen et al. [10] obtained arbitrary logarithmic savings for a specified class of fixed polynomial systems satisfying explicit degree–separation hypotheses, and Banks and Shparlinski [1] obtained logarithmic savings for sums with the factor $\mu(n_1n_2n_3)$ constrained by one equation $f(n_1) + g(n_2) + \wp(n_3) = M$, under their injectivity, bounded–weight, and scale hypotheses.

None of these theorems directly supplies the physical TPC estimate. The averaged Chowla theorem does not select the fixed shift h. Tao's two–point theorem does allow fixed affine forms, but its orbit average is logarithmically weighted and its coefficients are fixed as the asymptotic parameter tends to infinity; it supplies no uniformity for the X-dependent slopes m_α, m_β, nor for their coherent weighted summation. In the equality column those slopes satisfy

$$m_\alpha \asymp Q = J^{267/133+o(1)}, \tag{94}$$

and the target is a specific weighted quadratic form over many rows. The results on polynomial patterns use fixed systems and average a common translation. The Banks–Shparlinski theorem has one separable additive constraint and does not contain the two shared-j product slopes in (1). A theorem for each fixed row pair would also not automatically survive the coherent row summation at the required endpoint.

The exact kernel compression (84) suggests a different comparison: one now seeks Möbius cancellation against the packet–generated coefficient $C(\kappa)$. That coefficient is not presently known to belong to any coefficient class controlled by the cited theorems—for example, a fixed polynomial/nilsequence family or a family admitting a bounded–complexity separable additive representation. We therefore use the cited results only to mark the current boundary, not as black boxes proving (95).

10.3 Scope of the probabilistic language

"Annealed" refers to expectation over the prime signs while keeping the entire TPC packet fixed. "Almost surely along a prescribed dyadic sequence" refers to one global prime field and one fixed deterministic packet at each scale. No statement is made uniformly over all possible packets or about the distinguished all–minus point. The one–corner extremizer in theorem 9.1 shows that these qualifications cannot be dropped by measure theory alone.

11 The annealed–to–quenched gate

The preceding theorems separate two statements that were previously mixed together. The packet geometry is generically compatible with the endpoint: a single multiplicative prime field diagonalizes the equality alias exactly and controls the remaining aliases through subpolynomial product fibers, so almost every such field closes the full trace. The physical Möbius function is nevertheless one specified point of that field, not a random draw that may be selected after the energy is measured.

By theorem 8.1, the exact remaining equality theorem is the one–sided bound

$$\sum_{\kappa>1} \mu(\kappa)C(\kappa) \leq X^{o(1)} K_B Q^2 J, \tag{95}$$

with $C(\kappa)$ defined by the actual masks and row pairs in (82). The already controlled diagonal $\mathcal{D}_0$ is harmless at this scale. Formula (95) is a compression of the physical four–Möbius gate; it is not a known Möbius orthogonality theorem, because the coefficients aggregate growing affine slopes and may be highly correlated with κ.

Three precise continuations remain.

(a) **Rough–kernel descent.** Use corollary 8.2 to estimate the spectrum with $P^-(\kappa) > z$, then control the successive small–prime increments without paying their number absolutely.

(b) **Walsh spectral flatness.** Prove a coefficient estimate for $C(\kappa)$, such as a sufficiently strong weighted ℓ^2 or divisor–fiber bound, and combine it with a Möbius estimate adapted to the resulting kernel support. Parseval (85) measures the exact annealed variance, but by theorem 9.1 variance alone is insufficient.

(c) **Prime–phase de–randomization.** Prove directly that the all–minus phase cannot lie in the coherent exceptional set for the actual TPC coefficient system. Such a result must exploit arithmetic information beyond multiplicativity and squarefree support; an abstract selection from a measure–one set cannot identify a prescribed phase point.

The first route is now the most concrete next layer: it turns a four–sign row interaction into a single squarefree–kernel spectral sum and exhibits an exact prime filtration.

11.1 Claim boundary

This paper does not prove (95) at the physical point. It proves no fixed–shift Chowla estimate, no deterministic orbit–slope Bessel theorem for the Möbius coefficients, no parity breach, no Hardy–Littlewood prime–pair asymptotic, and no infinitude of twin primes. "Almost every prime–sign field" never means "the Möbius field": the latter is the distinguished all–minus point, and theorem 9.1 proves that this logical distinction is sharp.

12 Exact finite certificate and claim ledger

The companion script

`experiments/tpc43_certificate.py`

uses only Python's standard library and exact integer, rational, and Gaussian–integer arithmetic. It performs no random sampling. Its finite regressions verify:

(i) the squarefree–kernel character identity (39);

(ii) the divisor reconstruction and multiplicity bound in lemma 4.1;

(iii) zero–alias fused–label injectivity on exhaustive admissible toy packets;

(iv) exact Walsh second moments and the diagonal identity (62);

(v) the four–to–one identity (87);

(vi) the rough–kernel filtration (88);

(vii) exact Walsh Parseval for the grouped spectrum; and

(viii) the sharp one–corner extremizer in theorem 9.1.

The script also checks the endpoint exponent ledger and scans its own syntax tree to exclude floating–point literals, random imports, true division, and optimization–sensitive `assert` statements. The frozen run performs (43,877) exact checks. Its canonical certificate digest is

`d7d2b07107a6da1bd242ef9dbd8c02de7daea24dfdfbf9b5f31178cacf834d01`.

Normal and optimized Python executions produce the same JSON file, whose SHA–256 digest is

`f46e5b5ef49cc5f12f92ba0f702938a3fa2367b30c694e01f93fd14d30bd628a`.

Finite computation certifies the algebraic identities and finite combinatorics only. The asymptotic divisor bound, Bonami–Beckner inequality, inherited TPC diagonal estimates, and Borel–Cantelli lemma are mathematical inputs proved or cited in the text. The certificate does not test a physical Möbius asymptotic, affine Chowla cancellation, prime pairs, or twin primes.

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